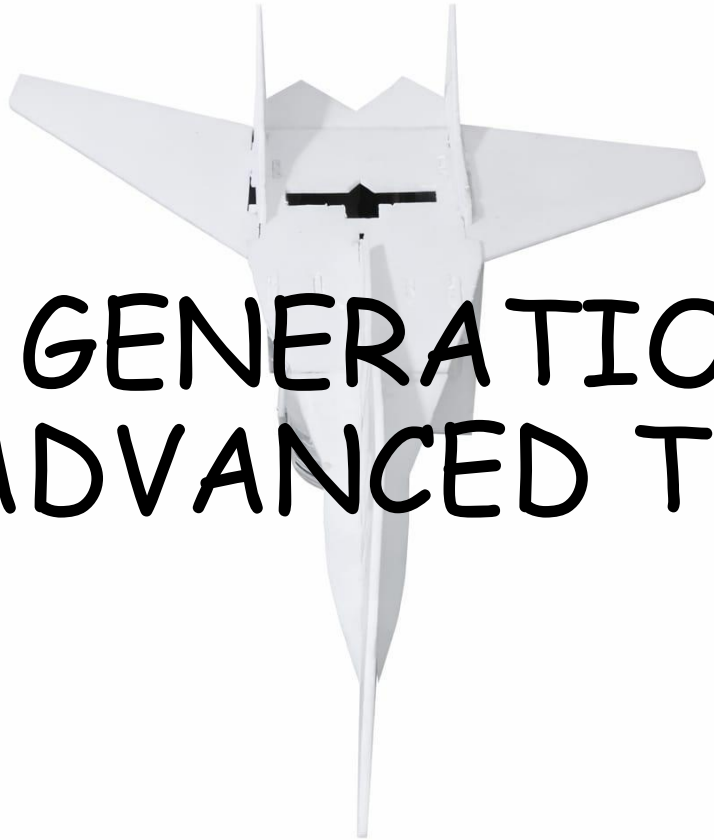


# MIG 29 5TH GENERATION FIGHTER JET WITH ADVANCED TECHNOLOGY

# GOVERNMENT POLYTECHNIC COLLEGE KANNUR

*ELECTRONICS SEMESTER 4*

**PARTICIPANTS**

ARSHIN PK

ASHWANTH K

ANJITH KK

DEVADETH KP

NIVED MO

SAYOOJ P

# INTRODUCTION

*Aircraft technology has developed greatly with the integration of electronics and automation systems. In this project, a small aircraft model is designed and controlled wirelessly using a transmitter and receiver system. The system uses an Arduino Nano as the main controller to manage the motor, servos, and communication modules. To improve safety during flight, IR sensing technology is incorporated to detect nearby obstacles and help the aircraft avoid collisions. By combining wireless control with sensor-based detection, the project demonstrates how modern electronics can enhance the performance, safety, and intelligence of small aerial systems.*

# working

- *The NRF24L01 module is used to transmit control signals wirelessly from the remote to the RC plane*
- *The BLDC motor is used to generate thrust. When electrical power is supplied through the ESC, the motor rotates the propeller and produces forward force.*
- *This thrust moves the RC plane forward. As the speed increases, air flows over the wings.*
- *The shape of the wings creates low pressure on the top side and high pressure on the bottom side of the wing.*
- *Due to this pressure difference, lift is generated, which allows the plane to rise into the air.*
- *Servo motors are connected to the control surfaces:*

*Elevator controls upward and downward movement*

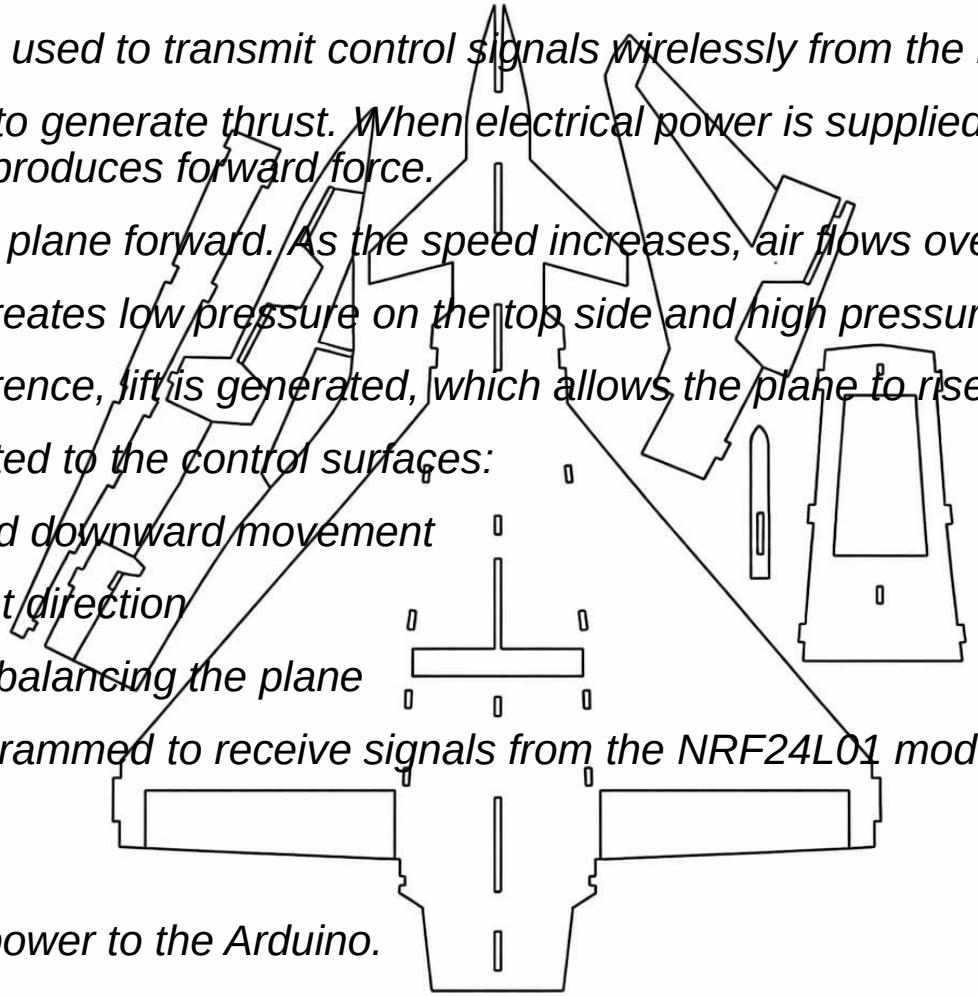
*Rudder controls left and right direction*

*Ailerons help in turning and balancing the plane*

- *The Arduino Nano is programmed to receive signals from the NRF24L01 module and control the ESC and servo motors accordingly.*

*Use C++ program*

- *A Li-ion battery supplies power to the Arduino.*



# components

- BLDC motor
- Receiver & transmitter
- Li-ion battery
- Joy stick
- Arduino Nano
- NRF24L07 wireless transceiver
- NRF2401 adapter board
- Potentiometer
- Thermo coil
- Servo
- ESC
- IR senser

# uses

- Obstacle Avoidance Training – Helps beginners fly RC planes safely by automatically avoiding obstacles.
- Surveillance – Can monitor areas while avoiding trees, buildings, or other objects.
- search and Rescue – Can fly in disaster areas and avoid debris or obstacles.
- Military Reconnaissance – Used for spying or monitoring enemy areas safely.
- Aerial Photography – Helps capture photos/videos without crashing into obstacles.
- Agricultural Monitoring – Can inspect crops while avoiding trees or structures.
- Educational Robotics Project – Demonstrates autonomous navigation and sensor technology.